A Policy Perspective on Electronic Commerce

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ABSTRACT The implementation of global electronic commerce is proceeding much more rapidly than initially anticipated. By 2003–2005, electronic commerce revenue is projected to total more than $1 trillion. The U. S. Department of Commerce's "The Emerging Digital Economy II" [1] provides a snapshot of how electronic commerce and information technologies are being used by businesses of all sizes and from all sectors to create new goods and services. Electronic commerce is changing the fundamental relationship between buyers and sellers, producers and consumers. The role of government in providing a policy framework deserves careful examination.

Anticipating the need to provide a policy basis for the growth of electronic commerce (EC), on July 1, 1997, President Clinton issued "A Framework for Global Electronic Commerce" [2, 3]. Because of its rapid development, and because the private sector was largely responsible for the commercialization of the Internet, the President identified the following policy principles:

- The private sector should lead.
- Governments should avoid undue restrictions on EC.
- Where governmental involvement is needed, its aim should be to support and enforce a predictable, minimalist, consistent and simple legal environment for commerce.
- Governments should recognize the unique qualities of the Internet.
- EC over the Internet should be facilitated on a global basis.

The last point was particularly important. The Internet and EC were born global. The President therefore set out a number of issues that had to be addressed in an international context if EC is to thrive.

Continuing the work of the Framework, the "U.S. Government Working Group on Electronic Commerce: First Annual Report" [4, 5] documents the U.S. progress in implementing the President's strategy and identifies five new areas that have emerged since the original Framework document was issued:

- Increasing the availability of bandwidth and Internet access in the United States
- Ensuring effective consumer protection in the online environment
- Expanding Internet availability and use of EC in developing countries
- Understanding better the economic impacts of the Internet and EC
- Facilitating small business and entrepreneurial use of the Internet and EC

In this article we discuss the work done in many of these policy areas as well as the activities of the G7/G8 Information Society Pilot Project on the Global Marketplace for Small- and Medium-Sized Enterprises (SMEs).

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THE ELECTRONIC COMMERCE POLICY INFRASTRUCTURE

Some policy areas might be thought of as providing an infrastructure, the absence of which would delay or hinder EC development. Some of these, discussed below, include access to bandwidth, the legal framework, security and authentication, taxation, tariffs and customs, electronic payment systems, reliability, universal access, and government support for R&D and standards.

Access to Bandwidth — For an EC participant to really enter cyberspace, for example, to browse in an electronic shop in a way that feels even somewhat similar to browsing in a local mall, they must depend on access to bandwidth. Although additional bandwidth for downloading is needed, today it is often the links to various servers that are the limiting factor. At the moment most people are connected to the Internet through dialup modems, with larger businesses having higher-bandwidth connections. Modem connections range from 28.8 to 56 kb/s. At the end of 1998, estimates indicated that there were on the order of 300,000–500,000 cable modems and perhaps 30,000 digital subscriber line (DSL) connections in the residential and small-business marketplace. While forecasts are for continued aggressive growth — up to 15 million households over the next three to five years — that would still represent only 15 percent of U.S. households. The Presidential directive of November 1998 [5] charged the Secretary of Commerce and the Federal Communications Commission (FCC) with expanding access to bandwidth in the United States. The FCC has responsibility, pursuant to Section 706 of the 1996 Telecommunications Act (www.fcc.gov/telecom.html), for looking at the state of access to advanced services across the country and ensuring that advanced services are universally available. There is considerable debate going on now as to how to facilitate the deployment of higher-bandwidth services. While this debate has largely been the province of telecommunications competitors, it is important to anyone interested in realizing the full potential of EC (see also the article by Rutkoski in this issue).

The President's directive [5] also suggested that it was important to improve access and Internet interconnectivity in other countries because this global medium will not thrive unless people around the world have access to the Internet at reasonable terms and conditions. The United States has supported activity in the World Trade Organization (WTO) (www.wto.org) over the last several years, and considerable progress has been made in opening telecommunications networks up to competition, but there is still a long way to go.

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It is very clear that the penetration rates in other countries have lagged behind that of the United States (see the article by O'Shea in this issue). It appears that major factors are the higher cost of Internet access due to the fact that billing is on a metered basis and the absence of a well-developed telecommunications infrastructure. In both of these cases, the United States believes that competition will lead to increased investment, the offering of new services, and a reduction in prices.

The Legal Framework — One of the principles in the Framework was that the legal structure be minimalist and predictable. For businesses, the predictability and enforceability of agreements, and knowledge of which conduct is acceptable and which prohibited are quite important. The contracts that businesses have with each other and with consumers in the physical world are usually required by law to be reduced to writing. But imagine the impact of requiring written paper-based contracts for every agreement on the Internet. The legal framework must change so that contracts formed electronically can be enforced in the courts. This requires changes in state and national laws, but is a necessary prerequisite for the growth of EC.

Security and Authentication — Connected to the legal issues surrounding contracts are issues of security and authentication, including the means by which people indicate assent to an agreement. Parties must feel confident about their knowledge of the person or persons with whom they are communicating electronically, certain that the information passed from party to party is not interfered with during the transmission, and secure that the means by which they provide assent to an agreement — such as a digital signature — can be enforced even if it is electronic (see the article by Wing and O'Higgins in this issue). These issues must be addressed by both technical and legal experts in order for EC to thrive. The U.S. government position is that the parties to an agreement should be able to choose the means for authentication, and that government policies should be technologically neutral given the rapidly developing technology of authentication.

The National Institute of Standards and Technology (NIST) is working closely with industry to develop computer security standards [6, 7]. For example, NIST is developing the Advanced Encryption Standard by sponsoring an open and international competition (http://csrc.nist.gov/encryption/aes). Also, NIST is developing, in partnership with interested industry via a Cooperative Research and Development Agreement (CRADA), the minimal interoperability specification for the exchange of digital certificates for the public key infrastructure (http://csrc.nist.gov/pki). Together with the Internet Engineering Task Force (IETF), NIST has developed the Internet Protocol security (IPsec) standard that allows the encryption of datagrams (http://ipsec-wit.antd.nist.gov). In addition to the above, NIST has been playing a proactive role in promoting the validation of the quality of commercial and secure cryptographic modules, firewalls, and related profiles via the only available open standard, the Common Criteria (http://niap.nist.gov).

Taxation — Like the enforcement of contracts, taxation is another area of governmental activity that potentially has a major impact on the development of EC. Last year there was a significant debate about taxation of Internet access and EC that led Congress to pass the Internet Tax Freedom Act in 1998 (www.house.gov/chriscox/nettax/lawsuml.html). This law established a three-year moratorium on taxation of Internet access and discriminatory taxes against EC, and established a commission that would look at issues involving remote sales: sales by catalog companies over the telephone, sales over the Internet, and treatment of similar sales by "brick and mortar" merchants. The Commission, made up of eight industry leaders and eight state and local governmental officials, along with the Secretaries of the Treasury and Commerce and the U. S. Trade Representative, is scheduled to report back to Congress by the spring of 2000. Other questions of taxation are being addressed internationally in the Organization for Economic Cooperation and Development (OECD) (www.oecd.org), which has been working hard to ensure nondiscriminatory treatment of EC internationally.

Tariffs and Customs — Another area that traditionally has been the province of government is that of tariffs and customs duties. The United States was able to obtain in the WTO a moratorium on any customs duties applied to the electronic delivery of any electronic information. Although there traditionally have been no duties on the delivery of faxes or e-mail, there have been duties on the physical media by which electronic information has been delivered (e.g., tapes and discs). There was a concern that as EC grew and potentially substituted for some forms of physical delivery of goods, people might move to a regime of imposing tariffs or duties on electronically delivered information. The United States is seeking to make the moratorium permanent in the WTO and is working to ensure that any trade-related regimes with respect to EC are the most liberal available to facilitate the growth of EC.

Electronic Payment Systems — EC has been greatly facilitated in the United States by the availability of credit card transactions for payments on the Internet. Governments around the world have an interest in ensuring that electronic payment systems are not easily susceptible to fraud and abuse, that they are not used for money-laundering purposes, and that their use does not destabilize the governments' currencies.

Security and Reliability of the Internet — Outages in the public switched network in the early 1990s led to the creation of the Network Reliability and Interoperability Council, which includes representatives of the operators of the public switched network and advises the FCC on steps to ensure the reliability and interoperability of the public switched network (www.nric.org). In the Internet space, there is no similar mechanism. However, recent outages in the electronic trading world, where over a third of retail security trades take place electronically, indicate that reliability may develop as an issue as EC becomes more and more central to the economic and social life of our country. In addition, the Department of Commerce has a responsibility, under the President's Directive concerning critical infrastructure protection, to ensure that the networks on which we all rely are secure.

Universal Access — The 1996 Telecommunications Act recognized for the first time in legislation the principle of universal service in the telephone world. The provisions of the Act recognized that all our citizens should have access to these essential capabilities in order to participate fully in our society. While household penetration of telephony hovers around 95 percent, penetration varies considerably among groups based on income, race, ethnicity, and geography.

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There is considerable disparity in access to the Internet along the same dimensions, as well as along the dimensions of gender and education. The Department of Commerce has studied this issue and has recently released the third report in its Falling Through the Net series "Defining the Digital Divide" [8]. As the Internet takes on a larger and larger role in our nation's life, we will have to address whether EC will thrive if we have a digital divide, a society with a large number of Internet haves and a considerable number of have-nots. To remedy the disparity, and to ensure that our young people have access to the skills necessary to participate in the digital economy, the Administration has sponsored a number of programs to ensure that schools, libraries, and community centers have access to the Internet.

The Role of Government in R&D and Standards — The government has been playing a key role in enabling EC markets by promoting infrastructural technologies, standards, and applications. Here, as in other areas, the mechanism has been to work closely with the private sector and academia. The Internet and Next-Generation Internet are well-known examples of such government technology policy, where the technologies that have been developed partly by government funds have been successfully commercialized. Furthermore, these efforts have produced useful de facto and open standards such as the Transmission Control Protocol/Internet Protocol (TCP/IP) and other Internet standards. In addition to the roles played by the Defense Advanced Research Projects Agency (DARPA), National Science Foundation (NSF), National Aeronautics and Space Agency (NASA), and Department of Energy (DOE) in promoting the Internet, NIST has been very active in promoting infrastructural standards for security, developing precompetitive technologies by sharing risk with the private sector, transferring technologies to SMEs, and participating in the making of software standards for middleware as well as manufacturing applications.

Some of the cooperative efforts between NIST and industry were included in the security section above. Other specific efforts in promoting infrastructural standards and technologies include the following. In partnership with formal standards bodies, consortia, and other fora, NIST participates in the development of middleware and manufacturing applications standards for software such as those for objects and for the exchange of product data; that is, the Standard for the Exchange of Product Model Data (STEP) (www.mel.nist.gov). In the area of technology development, the Advanced Technology Program (ATP) has been sharing risk with industry to develop technologies for healthcare, component-based software, and digital video (http://atp.nist.gov). With regard to technology transfer, the Manufacturing Engineering Partnership (MEP) partners with states and SMEs to promote manufacturing technology centers throughout the United States (http://www.mep.nist.gov).

Electronic Commerce Policies: The Applications Layer

If the policies surrounding bandwidth, standards, and so on might be thought of as infrastructural for EC, there are also important issues operating at what might be thought of as the application layer. Two critical issues are privacy and consumer protection.

The present high valuations of what are called ".com" companies depend on projections of continued rapid growth and vastly increased penetration of EC in both the business-to-business and business-to-consumer segments. This will not be true if buyers and sellers do not feel confident that the environment they are using is safe and secure. In addition to knowing with whom they are communicating and that their communications are secure, consumers need to know that personally identifiable information collected about them is not going to be misused, that they are protected from fraud and misrepresentation, and that they have some form of redress if an electronic transaction is not satisfactory.

Privacy — The Internet facilitates the collection of personally identifiable data. This data can be used for very positive purposes — to customize products, to provide better customer service, and to improve business processes — but it can also be used to invade the privacy of individuals. There is no other issue that so animates consumers. Of those not online, over 60 percent say they would be more likely to go online if their privacy were protected. Of those online and not purchasing, over 50 percent have concerns about unauthorized use of their personal data. According to another survey, the issues of privacy are more important for higher-income and higher-educated consumers than they are for lower-educated and lower-income consumers (www.privacyexchange.org/iss/surveys/surveys.html). A recent poll indicates that 58 percent of Internet users said they would be more likely to provide information to a site with a posted privacy policy and a seal from a well-known organization (www.research.att.com/library/trs/TRs/99/99.4/99.4.3/report.htm).

More and more companies are recognizing the need to develop and post strong privacy policies in order to increase consumer confidence in EC. The development of the Online Privacy Alliance (www.privacyalliance.org) over the last year demonstrated that many large EC players would be willing to post their privacy policies and have third-party firms such as TRUSTe (www.truste.com) and BBBOnLine (www.bbbonline.com) audit their implementation. IBM, Microsoft, and Disney's Buena Vista Internet-Intranet Group, among the largest Web advertisers, have recently committed to not advertising on sites that do not post clear privacy policies.

To increase consumer confidence that their information will not be abused, the privacy policies must be good ones. Over the years, the OECD has established a set of principles with respect to fair information practices. These include the right of consumers to know what information is being collected and how it will be used, the right of consumers to choose the information collected and the purposes for which it is used, the right of consumers to have access to the information to ensure it is accurate, and some enforceable means of legal redress for the misuse of the information. It will be important to follow developments in this area to see if companies undertake the development of policies that reflect these principles and enter into agreements with third parties to ensure that they are acting consistently with the policies. Similarly, it will be important to see what the government and the private sector may do to facilitate technological developments such as the Platform for Privacy Preferences (P3P) Project (www.w3.org/P3P), standards developed by the World Wide Web Consortium (W3C) that will provide individuals and companies effective and efficient means to determine the privacy preferences of individuals and offer appropriate privacy protections.

Consumer Protection — A similar situation exists with respect to consumer protection. Every state and every nation has a consumer protection law. This is not a new area for the government to enter. The question is, however, how does one have effective consumer protection laws in this electronic

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environment? Existing consumer protection rules from the physical world do apply in cyberspace but may need to be modified due to the unique qualities of the medium. Of particular concern is protecting against fraud and misrepresentation and providing appropriate means for the treatment of consumer complaints when providers are not in the same jurisdiction as the consumer. Imagine a situation with a company headquartered in California, a server located in New York, a customer in France, and the delivery of an electronic good to a recipient in India. Where does the transaction take place? Whose rules apply? How will the rules be enforced? These issues of jurisdiction, the right to choose which laws apply, the right to interpret those laws, and the right to enforce those laws, have very new challenges in EC. The Federal Trade Commission, working with the Department of Commerce, is having a series of workshops in this area (www.ftc.gov), while in the international arena, the OECD consumer protection committee is attempting to draw up consumer protection guidelines.

Content-Related Issues — The Internet has dramatically increased access to sources of information by users around the world, but at the same time this has created controversy about access to material that is deemed "harmful" such as pornography, hate speech, and information about building bombs. It is not surprising that other countries have differing views about what is harmful. Some have attempted to block content from abroad or even attempted to apply their criminal laws against Internet service providers or information providers who do not live or work in their countries. In general, the Administration has favored techniques for empowering users to better control their own communications through the availability of rating systems and filtering technologies. How we resolve these content issues must take into account the needs of special groups such as children, as well as the need to encourage a wide open and robust electronic marketplace of ideas.

Intellectual Property — Just as buyers seek the protection of effective consumer protection laws on the Internet, content creators need to feel confident about intellectual property protection for their material before they will make it available globally on a medium where instant and inexpensive copying is readily available. The WIPO treaties negotiated over the last several years are being implemented in the United States by two acts: the Digital Millennium Copyright Act [9] and the Millennium Digital Commerce Act [10]. This year Congress will resume its examination of database protection, attempting to resolve the tension between the rights of those people who compile them and the rights of people who have access to them.

Regulation of "Like" Services: Telephony, Broadcasting, and So On — As convergence becomes real with voice, data, image, and video being represented by ones and zeros, and as the Internet becomes a pervasive medium for the delivery of these bits, the issue arises of how to regulate services which previously have been provided over discrete media: copper pair for voice telephony, coaxial cable or over-the-air broadcasting for video, and so on. Should Internet telephony be regulated like the telephony provided by incumbent local telephone companies? Should streaming video or audio be regulated like cable TV service or like over-the-air radio or television broadcasting? Should a new regime be created, or should the existing regulatory schemes be employed, modified, or junked? Our public policy agenda must reflect the actuality of convergence and the central role played by the Internet.

SMEs and Electronic Commerce

The President directed the Small Business Administrator and the Secretary of Commerce to create a strategy which would ensure that our most vigorous sector, the small business sector, will be able to enjoy the fruits of EC. This activity will build on work undertaken in the G8 Pilot "Global Marketplace for SMEs" led by the European Commission, Japan, and the United States. While initially involving only the G7 countries and the European Commission, participation expanded to include over 20 countries and several international organizations, and contributed substantially to EC policy and actions in most of the participating countries.

The pilot project resulted in three areas of cooperative effort. First, a large network of business information on the Web was established, the Global Information Network for SMEs, with contributions from 15 economies and five international organizations (www.gin.smc.ne.jp). Second, international and national working groups studied the policy issues in EC for SMEs (legal, technical, institutional). This work influenced policy definitions in many countries and resulted in a best-practice book with examples of how SMEs deal with issues (www.martech-intl.com/best2/index.htm). Third, a registry of international testbeds or pilots that have received the G8 label was created. The United States served as registrar and established the Web site (http://nii.nist.gov/g7/10_global_mp/testbeds/registered.html) containing the testbed descriptions in a common format and providing links to the pilot projects' Web sites.

Major conferences of the project were held in April 1997, September 1998, and April 1999. Attendees expressed strong support for the work because it provided the only private/public sector forum where "policy meets practice," that is, where the practical implementation of EC tools and policy affecting SMEs is its focus. The final report of the pilot project recommends that future work should aim to:

- Provide information and improve awareness in training about EC aimed at SMEs, including continuation of the Global Information Network and the Global Marketplace Web site
- Promote international pilots for SMEs as participants in global trade and enable SMEs to make the transition from the paper-based economy to the digital economy
- Strengthen international collaboration between emerging, developing, and industrialized economies in EC for SMEs [11].

Conclusion

In 1997 the United States made public its Framework for EC. Work is progressing following the guidelines established by that document, and taking into consideration global efforts and new requirements. It is increasingly clear that the Internet and EC will be a major force driving economic growth over the next few decades. The U.S. government is committed to implementing market-driven policies that encourage the development of this new digital economy in an environment that respects the concerns of individuals and families [4]. It is hoped that all governments will act in this manner to encourage the unfettered growth of global EC.

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